

FLUID-ICU data dictionary

> **SIMULATED DATA - NOT REAL PATIENTS.** FLUID-ICU was generated by R/00_generate_fluid_icu.R for teaching.

Study: FLUID-ICU, a multicentre prospective cohort of adults with sepsis admitted to 4 ICUs (ICU A-D), January 2023 to December 2024.

Files: fluid_icu_raw.csv / .xlsx (as sent by the sites, 623 rows, used in the cleaning demo) and fluid_icu_clean.csv / .rds (620 patients, 32 variables, analysis-ready).

Main exposure: cumulative fluid balance at day 3 (cum_fluid_balance_d3_ml_kg, positive_balance_d3). **Primary outcome:** died_28d (with time_to_event_28d).

Pre-specified confounders: age, sofa, lactate_mmol_l, vasopressor, mech_vent, ckd. **Effect modifier:** heart_failure.

In the clean file a missing value is an empty cell (CSV) or NA (R). *Missing code (raw)* gives how the raw file recorded a missing value. *Allowed range* is the plausibility check used in R/01_clean.R; values outside it are set to missing and queried. DERIVED variables are computed in cleaning and do not exist in the raw file.

#	Variable	Label	Type	Units / codes	Allowed range	Missing code (raw)	Notes
1	study_id	Study identifier	Character (ID)	FLU-001 ... FLU-620	Unique	None allowed	Raw column 'Study ID'. Raw file has 3 IDs entered twice (exact duplicate rows); removed in cleaning.
2	site	Recruiting ICU	Categorical (4 levels)	ICU A / ICU B / ICU C / ICU D	4 levels	None allowed	Raw 'Site' typed several ways (icu a, ICU-A, ICUA); standardised. ICU C reported lactate in mg/dL.
3	admission_date	Date of ICU admission (enrolment)	Date	YYYY-MM-DD	2023-01-01 to 2024-12-31	None allowed	Raw 'Admission Date' mixes YYYY-MM-DD, DD/MM/YYYY and DD-Mon-YYYY.

#	Variable	Label	Type	Units / codes	Allowed range	Missing code (raw)	Notes
4	age	Age	Numeric (integer)	Years	18 to 110	Blank	Raw 'Age'. One impossible value (210) set to missing; query to site.
5	sex	Sex	Categorical (2 levels)	Female / Male	2 levels	None allowed	Raw 'Sex' has 8-9 spellings (M, m, male, F, f, female, 'Female').
6	weight_kg	Body weight on admission	Numeric	kg (1 decimal)	30 to 250	Blank	Raw 'Weight (kg)'. One impossible value (7) set to missing. Used to compute balance in ml/kg.
7	diabetes	Diabetes mellitus	Binary	No / Yes	No, Yes	None	Raw codes Yes/No/Y/N /yes/no/1/0 (ICU B sends 1/0).
8	ckd	Chronic kidney disease	Binary	No / Yes	No, Yes	None	Pre-specified confounder. Mixed Yes/No coding in raw file.
9	heart_failure	Chronic heart failure	Binary	No / Yes	No, Yes	None	Pre-specified effect modifier (Q1). Mixed coding in raw file.
10	hypertension	Hypertension	Binary	No / Yes	No, Yes	None	Mixed coding in raw file.
11	infection_source	Source of infection	Categorical (5 levels)	Respiratory / Urinary / Abdominal / Skin/soft tissue / Other	5 levels	None	Ordered as listed for tables.

#	Variable	Label	Type	Units / codes	Allowed range	Missing code (raw)	Notes
12	sofa	SOFA score on ICU admission	Numeric (integer score)	Points	0 to 24	None	Severity; pre-specified confounder. Median 7 (IQR 5-9).
13	apache_ii	APACHE II score (first 24 h)	Numeric (integer score)	Points	0 to 71	-99	Raw 'APACHE II'; 6 values coded -99 = not recorded.
14	lactate_mmol_l	Blood lactate on admission	Numeric	mmol/L (1 decimal)	0.1 to 30	Blank, 999, -99	~15% missing, more often in less severe patients (MAR). ICU C values in raw file are mg/dL (x 9.01): divided by 9.01 in cleaning.
15	map_mmhg	Mean arterial pressure on admission	Numeric (integer)	mmHg	20 to 200	999	Raw 'MAP (mmHg)'; 4 values coded 999.
16	vasopressor	Vasopressor on admission	Binary	No / Yes	No, Yes	None	Pre-specified confounder. Mixed coding in raw file.
17	mech_vent	Invasive mechanical ventilation on admission	Binary	No / Yes	No, Yes	None	Raw 'Mech vent'. Pre-specified confounder.
18	fluid_in_d1_ml	Total fluid input, ICU day 1	Numeric (integer)	ml	0 to 20000	Blank	All intravenous and enteral fluid incl. drugs and nutrition. Raw 'Fluid in D1 (ml)'.
19	fluid_output_d1_ml	Total fluid output, ICU day 1	Numeric (integer)	ml	0 to 20000	Blank	Urine, drains, ultra filtration; insensible losses not included.

#	Variable	Label	Type	Units / codes	Allowed range	Missing code (raw)	Notes
20	fluid_in_d2_ml	Total fluid input, ICU day 2	Numeric (integer)	ml	0 to 20000	Blank	
21	fluid_output_d2_ml	Total fluid output, ICU day 2	Numeric (integer)	ml	0 to 20000	Blank	One negative value (impossible) in raw file set to missing.
22	fluid_in_d3_ml	Total fluid input, ICU day 3	Numeric (integer)	ml	0 to 20000	Blank	
23	fluid_output_d3_ml	Total fluid output, ICU day 3	Numeric (integer)	ml	0 to 20000	Blank	
24	cum_fluid_balance_d3_ml_kg	Cumulative fluid balance, ICU days 1-3	Numeric (DERIVED)	ml/kg (1 decimal)	-150 to 250	Derived	= (sum of fluid in d1-3 - sum of fluid out d1-3) / weight_kg. Not in the raw file. Missing for 2 patients (weight or output missing). Main exposure (Q2, Q3: per 10 ml/kg).
25	positive_balance_d3	Positive cumulative fluid balance at day 3	Binary (DERIVED)	No / Yes	No, Yes	Derived	Yes if cum_fluid_balance_d3_ml_kg > 0. Main exposure for Q1 and Q5.
26	aki_stage	Acute kidney injury, maximum KDIGO stage, ICU days 1-7	Ordinal (integer)	0 = none, 1, 2, 3	0 to 3	None	Raw 'AKI stage (KDIGO)'.
27	aki_any	Any acute kidney injury (KDIGO stage >= 1)	Binary (DERIVED)	No / Yes	No, Yes	Derived	Yes if aki_stage >= 1. Outcome for Q2.

#	Variable	Label	Type	Units / codes	Allowed range	Missing code (raw)	Notes
28	rrt	Renal replacement therapy by day 28	Binary	No / Yes	No, Yes	None	Mixed coding in raw file.
29	vent_free_days_28	Ventilator-free days to day 28	Numeric (integer)	Days	0 to 28	None	Days alive and free of invasive ventilation to day 28; 0 if died; 28 if never ventilated and alive. Raw 'VFD 28'. Outcome for Q3.
30	icu_los_days	ICU length of stay	Numeric	Days (1 decimal)	> 0	None	Right-skewed. For patients who died, stay ends at death. Outcome for Q5 (survivors).
31	died_28d	Death from any cause by day 28	Binary	No / Yes	No, Yes	None	Primary outcome (Q1, Q4). Mixed coding in raw file.
32	time_to_event_28d	Time from ICU admission to death or censoring	Numeric (integer)	Days	1 to 28	None	Death day if died; 28 (censored) if alive at day 28. Use with died_28d for Kaplan-Meier and Cox. Raw 'Time to event 28d (days)'.

Column names in the raw file are spreadsheet-style (e.g. Weight (kg), MAP (mmHg));
 janitor::clean_names() plus four renames give the names above (see R/01_clean.R, step 1).